

eBay E-Commerce Sales Dashboard

Project Overview — Data Errors, Cleaning Steps & Excel Solutions

1. Project Overview

This project analyzes eBay e-commerce sales data consisting of 13,000 transactional records across 5 product categories and 12 major US cities. The raw dataset was intentionally messy — containing mixed date formats, inconsistent customer names, NULL values, duplicate records, and misformatted prices. Using only Microsoft Excel, the data was cleaned, structured, and transformed into an interactive executive dashboard with PivotTables, advanced formulas, KPI cards, and connected slicers.

Tool Used	Dataset	Records	Categories	Cities	Year
Microsoft Excel	eBay Sales CSV	13,000 Rows	5 Product Lines	12 US Cities	2024

2. Raw Data Problems Found (What Was Wrong)

The Raw-Data.csv contained **8 categories of data quality issues** across 13,000 rows. Each problem is documented below with an example of what the messy data looked like.

1	Mixed Date Formats		
Problem: Same date column had 4 different formats mixed together — impossible to sort or calculate.			
Raw (Messy)		Fixed (Clean)	
1/15/25		2024-01-15	
01-15-2024		2024-01-15	
2024/01/15		2024-01-15	
15/01/24		2024-01-15	
2	Inconsistent Customer Names + City Suffix		
Problem: Customer names included city codes appended with a dash — needed to be split into separate columns.			
Raw (Messy)		Customer (Clean)	City (Clean)
"Alice - NYC"		Alice	NYC
"Bob - LA"		Bob	LA
"Charlie - CHI"		Charlie	CHI
3	Mixed Price Formats		
Problem: Price column had 4 different formats — some with \$ symbol, some missing decimals, some abbreviated.			
Raw (Messy)		Fixed (Clean)	
\$34.99		34.99	
35		35.00	
34.9		34.90	
N/A		0.00 (IFERROR)	

4 Inconsistent Status Casing

Problem: Order status had the same values written in different cases — treated as different categories by Excel.

Raw (Messy)	Fixed (Clean)
delivered	Valid
Delivered	Valid
RETURNED	Invalid
shipped	Invalid
processing	Invalid

5 Mixed OrderID Formats

Problem: OrderID column mixed plain numbers with prefixed text strings — inconsistent for lookups.

Raw (Messy)	Fixed (Clean)
1001	1001
ORD-1002	1002
ORD-1003	1003

6 Missing / NULL Values

Problem: Approximately 200 rows had blank Qty, N/A prices, or empty Status fields injected as dirty records.

Column	Issue	Fix Applied
Qty	Blank cells	=IFERROR(A2,0)
Price	N/A text	=IFERROR(A2,0)
Status	Empty cells	=IF(A2="", 'Invalid', A2)

7 Duplicate Records

Problem: Some OrderID + Date combinations appeared more than once — inflating revenue and order counts.

Raw (Messy)	Action Taken
1001 2024-01-15 x2	Removed duplicate via Data → Remove Duplicates
1050 2024-03-10 x3	Kept first occurrence only

8 Quoted Product Names

Problem: Some product names were wrapped in extra quotation marks causing mismatches in PivotTables.

Raw (Messy)	Fixed (Clean)
"'T-Shirt'"	T-Shirt
"'Desk Lamp'"	Desk Lamp

3. Step-by-Step Cleaning Process (Excel Solutions)

Every data problem was solved using native Excel tools and formulas — no Python, no macros. Below are the exact steps taken in order, with the formula or Excel feature used.

Step 1 Split Customer Name + City

Method :	Data tab → Text to Columns → Delimited → Dash (-) separator
Formula:	<code>=TRIM(LEFT(A2, FIND("-",A2)-1))</code> → Customer Name <code>=TRIM(MID(A2, FIND("-",A2)+1, 100))</code> → City
Result:	Customer and City now in separate columns. Trailing spaces removed with TRIM().
Step 2 Standardize All Dates	
Method :	All 4 date format variations converted to a single consistent YYYY-MM-DD format
Formula:	<code>=DATEVALUE(TEXT(A2,"YYYY-MM-DD"))</code> then Paste Special → Values Only
Result:	Dates now sortable and usable in SUMIFS and PivotTables. Stored as Excel date serial numbers.
Step 3 Remove \$ Symbol from Prices	
Method :	Find & Replace (Ctrl+H) → Find '\$' → Replace with " (blank)
Formula:	<code>=VALUE(SUBSTITUTE(G2,"\$",""))</code> as backup formula for stubborn cells
Result:	All prices now stored as pure numbers. Formatted as Currency in column.
Step 4 Standardize Order Status	
Method :	Used IF formula to collapse all status variations into two clean values: Valid / Invalid
Formula:	<code>=IF(LOWER(I2)="delivered", "Valid", "Invalid")</code>
Result:	Column I now contains only 'Valid' or 'Invalid'. Used in SUMIFS for dashboard KPIs.
Step 5 Fix OrderID Format	
Method :	Used SUBSTITUTE to remove 'ORD-' prefix from text-based IDs, then converted to number
Formula:	<code>=VALUE(SUBSTITUTE(A2,"ORD-",""))</code>
Result:	All OrderIDs now consistent numeric format. Enables proper deduplication.
Step 6 Handle NULL / Missing Values	
Method :	Wrapped all formula cells with IFERROR to catch blank, N/A, and error values
Formula:	<code>=IFERROR(VALUE(G2), 0)</code> → for Price <code>=IF(F2="", 1, F2)</code> → for Qty blank fallback
Result:	No more #VALUE! or #N/A errors. Blank cells replaced with 0 to prevent formula breaks.
Step 7 Remove Duplicate Records	
Method :	Data tab → Remove Duplicates → Selected columns: Order_id + Date

Formula:	Built-in Excel feature – no formula needed
Result:	Duplicate rows removed. Each Order_id + Date combination now appears only once.
Step 8	Create Revenue Column
Method:	Multiplied Qty × Price to get per-row revenue. This feeds all PivotTables and KPIs.
Formula:	=F2*G2 → Revenue (new Column J)
Result:	Revenue column created for all 13,000 rows. Used as primary metric across entire dashboard.
Step 9	Create Month Column
Method:	Extracted Month-Year label from Date column for use in Monthly Trends PivotTable
Formula:	=TEXT(D2, "MMM-YY") → e.g. Jan-24, Feb-24
Result:	Month column enables grouping in PivotTable 2 (Monthly Trends) and Product Matrix.
Step 10	Customer Tier Classification
Method:	Used SUMIF to calculate each customer's lifetime revenue, then INDEX-MATCH to assign tier
Formula:	=SUMIF(\$B:\$B,B2,\$J:\$J) → Customer Total Revenue (Col K) =INDEX(TierTable,MATCH(K2,Thresholds,1)) → Bronze/Silver/Gold (Col L)
Result:	Tier thresholds set at 0 / 270,000 / 310,000 based on actual data range (\$242K–\$340K).

4. Before vs. After — Data Transformation

A side-by-side comparison of the raw messy data versus the cleaned production-ready dataset:

BEFORE — Raw Messy Data (CSV)										
OrderID	Customer	Date	Product	Category	Qty	Price	Status			
ORD-1001	"Alice - NYC"	1/15/25	Laptop	Electronics	2	\$50.00	delivered			
1002	"Bob - LA"	01-20-2025	"T-Shirt"	Clothing		25	RETURNED			
ORD-1003	"Charlie-CHI"	2025/01/25	Mouse	Electronics	3	N/A	Delivered			
1004	"Diana - NYC"	25/01/24	Desk Lamp	Office	2	\$34.99	shipped			
AFTER — Clean Production-Ready Data										
Order_id	Customer	City	Date	Product	Qty	Price	Status	Revenue	Month	Tier
1001	Alice	NYC	2024-01-15	Laptop	2	50.00	Valid	100.00	Jan-24	Gold
1002	Bob	LA	2024-01-20	T-Shirt	1	25.00	Invalid	25.00	Jan-24	Silver

1003	Charlie	CHI	2024-01-2	Mouse	3	15.00	Valid	45.00	Jan-24	Gold
			5							
1004	Diana	NYC	2024-01-2	Desk Lamp	2	34.99	Invalid	69.98	Jan-24	Silver

5. New Columns Created During Cleaning

New Column	Formula Used	Purpose
Revenue	=Qty × Price	Core metric for all PivotTables and KPI calculations
Month	=TEXT(Date, "MMM-YY")	Used in Monthly Trends PivotTable and Product Matrix
Status (clean)	=IF(LOWER(raw)="delivered", "Valid", "Invalid")	Normalizes 5 status variants into 2 clean values
City	Text to Columns split from Customer	Enables city-level filtering via slicers
Cust Total Revenue	=SUMIF(Customer col, Customer, Revenue)	Lifetime revenue per customer — feeds tier logic
Cust Tier	=INDEX-MATCH on Tier Table	Bronze / Silver / Gold based on lifetime revenue

6. Final KPI Results After Cleaning

TOTAL REVENUE	TOTAL ORDERS	AVG ORDER VALUE	MoM GROWTH	TOP CATEGORY
\$5,744,894	13,000	\$457.97	+44.2%	Electronics
Full Year 2024	Validated rows	Per transaction	\$93K → \$134K	59% of Revenue

7. PivotTables Built for Analysis

PivotTable 1	Revenue by Category		
	Category	Count of Orders	Sum of Revenue
	Electronics	2,513	\$3,378,103
	Clothing	2,607	\$632,782
	Office	2,616	\$633,753
	Home	2,594	\$620,761
	Sports	2,670	\$479,493
PivotTable 2	Monthly Revenue Trends (Jan–Dec 2024)		
	Month	Orders	Revenue
	Jan-24	1,120	\$458,226
	Feb-24	1,049	\$486,664
	Mar-24	1,147	\$534,750
	Apr-24	1,053	\$477,054
	Jul-24	1,135	\$503,891
	Dec-24	1,071	\$416,391

PivotTable 3	Top 10 Customers by Revenue		
Customer		Total Revenue	Tier
Charlie		\$340,431	Gold
Ivy		\$339,833	Gold
Tina		\$322,714	Gold
Paul		\$316,435	Gold
Eve		\$308,798	Silver

PivotTable 4	Product Matrix — Category × Month Heatmap			
Month	Electronics	Clothing	Home	Office
Mar-24	\$318,686	\$55,120	\$52,721	\$61,529
Jul-24	\$279,878	\$56,871	\$59,373	\$52,737
Sep-24	\$329,630	\$43,790	\$46,443	\$48,733
Dec-24	\$241,372	\$43,732	\$45,922	\$47,756

8. Customer Tier Classification System

Since customer lifetime revenues ranged from \$242,560 to \$340,431 — a narrow \$97K band — standard thresholds (0/100/500) were irrelevant. Thresholds were set using the actual data distribution to create three meaningful segments across the real revenue range.

Tier	Threshold	Revenue Range	Formula
■ Bronze	0	\$242,560 – \$269,999	=INDEX(TierTable,MATCH(K2,Threshol ds,1))
■ Silver	270,000	\$270,000 – \$309,999	Same INDEX-MATCH — position 2
■ Gold	310,000	\$310,000 – \$340,431	Same INDEX-MATCH — position 3